



TECHNICAL REPORT

A Psychrometric Temperature Correction for the Positive Bias Observed in Stomatal Conductance Measured by the Open Flow-Through LI-600 Porometer

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ABSTRACT

The development of commercially available porometers has allowed for higher throughput measurement of stomatal conductance, but a body of evidence has suggested a persistent positive bias in their measurements relative to “reference” measurements from instrumentation based on infra-red gas analysis. We compiled a data set comprised of 25 angiosperm species, across a range of field conditions and found that the LI-COR LI-600, an open flow-through porometer, produced an exponentially increasing bias relative to the LI-COR LI-6800 infra-red gas analyser-based instrument in response to increasing stomatal conductance and decreasing relative humidity. This bias was minimal at lower stomatal conductance (below roughly $0.25 \text{ mol m}^{-2} \text{ s}^{-1}$), but was pronounced for larger values. We hypothesised that this bias is the result of the assumption of a constant air temperature throughout the flow stream used by the instrument software to estimate stomatal conductance from raw sensor measurements. We relaxed this assumption, and applied psychrometrics to augment the typical gas exchange equations with an additional energy balance constraint to solve for the temperature change throughout the air flow stream. We found that including this temperature difference corrects the computed transpiration and stomatal conductance values, and brings the porometer measurement into agreement with that of the infra-red gas analysis-based system. Software is provided to apply the correction to LI-600 output files. For future instrument design iterations, explicit measurement of temperature variation in the flow stream provides a potential opportunity for improvement in measurement accuracy at high stomatal conductance.

1 | Introduction

Stomatal conductance is a critical regulator of terrestrial carbon and water fluxes, yet accurate measurement of stomatal conductance for high-throughput agricultural and large-scale ecophysiological studies remains challenging. Leaf-level stomatal conductance to water vapour (g_{sw}) is primarily measured with one of two instruments: porometers or portable gas exchange

systems equipped with infra-red gas analysers (IRGAs). Trade-offs exist between the two instruments regarding precision, portability, efficiency, and expense, with porometers generally being less accurate but more portable, faster, and less expensive. Challenges associated with the reliability of porometer measurements have been identified and discussed in previous work (Turner 1991), particularly under high chamber relative humidity (McDermitt 1990), large gradients in leaf-to-air

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temperature (Tyree and Wilmot 1990), and large gradients in leaf-to-sensor housing temperature (Verhoef 1997). These factors have been reported to result in up to a 50% positive bias in measured stomatal conductance values over their “reference” IRGA-based counterparts (Lavoie-Lamoureux et al. 2017; Toro et al. 2019).

This measurement bias may not be problematic for comparative studies where only relative changes in stomatal conductance are needed, but when the absolute magnitude of stomatal conductance is important, a 50% bias is unacceptable. As modelling continues to play an increasingly important role in agriculture, ecology, and climate science, stomatal conductance becomes an increasingly essential model component to accurately quantify in absolute terms. Yield predictions, wildfire fuel forecasts, and global temperature projections are currently and will continue to be influenced by the absolute magnitude of stomatal conductance provided by porometer measurements (e.g., Lu et al. 1998; Yang et al. 1998; Wullschlegel et al. 2002; Nolan et al. 2018). The high throughput nature of modern porometers offers a promising solution to the slower and more expensive instrument alternatives (Chaplin et al. 2024), but if their absolute accuracy is unreliable then this presents a problem for inter-experiment comparisons and for their use in integrative modelling research that attempts to scale from leaf to canopy to globe (Schulze et al. 1994; Roy et al. 2012; Franks et al. 2018).

In this study, we explored the problem of bias in porometry-based measurement of stomatal conductance, focusing on the most recent generation of porometer instruments utilising an open flow-through design. While previous work has also reported a positive bias in diffusion-based porometers due to instrument temperature gradients as introduced above (Lavoie-Lamoureux et al. 2017; Toro et al. 2019), the underlying physical basis of this older class of instrument is entirely different from the open flow-through design considered herein. Thus, the discussion and results below are strictly restricted to open flow-through designs. We collected a novel set of stomatal conductance measurements across a diverse set of species and compiled them alongside a previously published data set (Kimura et al. 2025) to quantify discrepancies between the LI-COR LI-600 porometer and LI-6800 portable gas exchange system (LI-COR Biosciences, Lincoln, Nebraska, USA). We hypothesised that the assumption of constant air temperature throughout the flow stream made in the LI-600 processing software could cause a substantial bias in estimated stomatal conductance values. The goal of this study was to determine whether this assumption could explain the bias in observed measurements, and if true, whether this bias could be corrected by estimating temperature variation in the flow stream.

2 | Theory

In this study, we focus on steady-state, open flow-through porometers. This type of porometer utilises a steady flow of air passing over a leaf sample segment contained within a chamber (Figure 1). By measuring the change in relative humidity of the air stream as it passes through the chamber (along with other ancillary measurements), the amount of water vapour added to the air stream via leaf transpiration can be estimated. From this

and ancillary measurements, stomatal conductance can be inferred.

All constants and variables used in the following derivation are listed in Table A1.

The flux of water vapour at the leaf surface inside the chamber, E ($\text{mol m}^{-2} \text{s}^{-1}$), is driven by the water vapour mole fraction difference between the leaf interior and ambient chamber air, and is modulated by the conductance to water vapour between the leaf interior and outside of the leaf boundary-layer. This can be expressed as

$$E = g_{nw} \frac{e_s(T_{\text{leaf}})RH_{\text{leaf}} - e_s(T_{a,\text{cham}})RH_{\text{cham}}}{P_{\text{atm}}}, \quad (1)$$

where g_{nw} ($\text{mol m}^{-2} \text{s}^{-1}$) is the conductance to water vapour transfer between the leaf interior and outside of the leaf boundary-layer, $e_s(T)$ (kPa) is the saturation vapour pressure at temperature T , T_{leaf} is the temperature of the leaf surface ($^{\circ}\text{C}$), $T_{a,\text{cham}}$ is the ambient chamber air temperature ($^{\circ}\text{C}$), RH_{leaf}

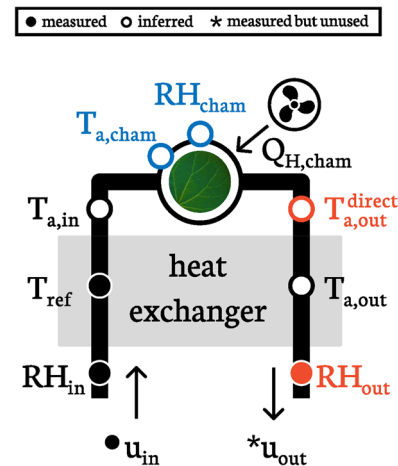


FIGURE 1 | Schematic representation of the LI-600 porometer flow path and the variables needed to compute leaf transpiration rate (E) and stomatal conductance (g_{sw}). Directly measured variables appear with filled dots, and variables not measured but inferred based on assumptions appear with open dots. The outlet flow rate, u_{out} , appears with an asterisk because while it is directly measured, its accuracy is not sufficient to be useful in calculations. Air temperatures, $T_{a,x}$, and relative humidities, RH_x , at different locations, x , along the flow path are coloured according to assumed equality. Leaf temperature, T_{leaf} (omitted for figure clarity) is measured with an infra-red thermometer pointed at the leaf surface. The LI-600 was designed to minimise air temperature gradients along the flow path by passing air through an aluminium block that acts as a heat exchanger, and by using internal fans to quickly mix air. The key assumption made by the LI-600's internal calculations for E and g_{sw} is that $T_{a,in}$, $T_{a,out}^{\text{direct}}$, and $T_{a,out}$ are equal to T_{ref} , the temperature of a thermistor embedded in the aluminium block. This simplifies calculations of E and g_{sw} but is the cause of observed overestimation in g_{sw} , we hypothesise, because it ignores the heat addition by the fans, $Q_{H,\text{cham}}$, and does not account for differences between the direct chamber outlet air temperature $T_{a,out}^{\text{direct}}$, the diffused outlet air temperature $T_{a,out}$, and the reference block temperature T_{ref} . [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/pce.70304)]

(unitless) is the relative humidity of the leaf internal air space, assumed to be 1.0, RH_{cham} is the ambient chamber relative humidity (unitless), and P_{atm} is the ambient chamber air pressure (kPa). $e_s(T)$ can be calculated using the Tetens equation

$$e_s(T) = 0.61365 \exp\left(\frac{17.502 T}{240.97 + T}\right), \quad (2)$$

where T is temperature in units of Celsius, and the resulting vapour pressure e_s is in units of kPa (Tetens 1930).

The overall conductance to water vapour, g_{sw} , is the combination of stomatal (g_{sw} ; $\text{mol m}^{-2} \text{s}^{-1}$) and boundary-layer (g_{blw} ; $\text{mol m}^{-2} \text{s}^{-1}$) conductances in series, ignoring cuticular conductance. The stomatal conductance can thus be expressed as

$$g_{\text{sw}} = \left(\frac{1}{g_{\text{sw}}} - \frac{1}{g_{\text{blw}}} \right)^{-1}. \quad (3)$$

Combining Equations (1) and (3) and solving for g_{sw} yields

$$g_{\text{sw}} = \left(\frac{e_s(T_{\text{leaf}}) - e_s(T_{\text{a,cham}})RH_{\text{cham}}}{EP_{\text{atm}}} - \frac{1}{g_{\text{blw}}} \right)^{-1}. \quad (4)$$

Assuming that we knew the temperature and humidity conditions in the chamber, the main unknown to be determined in the calculation of g_{sw} is the evaporation rate E . This can be determined through a moist air mass balance between the chamber input and output:

$$sE = u_{\text{out}} \frac{e_s(T_{\text{a,out}})RH_{\text{out}}}{P_{\text{atm}}} - u_{\text{in}} \frac{e_s(T_{\text{a,in}})RH_{\text{in}}}{P_{\text{atm}}}, \quad (5)$$

where s is the one-sided leaf area exposed in the chamber (m^2), u_{in} and u_{out} are respectively the molar mass flow rates of moist air into and out of the chamber (mol s^{-1}), RH_{in} is the inlet relative humidity, $T_{\text{a,in}}$ is the air temperature of the flow coming into the chamber, and $T_{\text{a,out}}$ is the air temperature of the flow coming out of the chamber after it has passed through the heat exchange block, at the location of the outlet relative humidity sensor for RH_{out} (see Figure 1). It is important to note that the flow rates are given in units of moles of moist air per second, and thus the vapour fraction is given in units of moles of water vapour per mole of moist air, and therefore u_{in} is not equal to u_{out} as is usually the case when flow rates are given in terms of moles of dry air. If u_{out} is not known, it can be determined by assuming that the only change in moist air mass is the addition of water vapour to the incoming air stream from the transpiring leaf:

$$u_{\text{out}} = u_{\text{in}} + sE. \quad (6)$$

Substituting Equation (6) into Equation (5) and solving for E yields:

$$E = \frac{u_{\text{in}}}{s} \left(\frac{e_s(T_{\text{a,out}})RH_{\text{out}}}{P_{\text{atm}}} - \frac{e_s(T_{\text{a,in}})RH_{\text{in}}}{P_{\text{atm}}} \right) \left(1 - \frac{e_s(T_{\text{a,out}})RH_{\text{out}}}{P_{\text{atm}}} \right)^{-1}. \quad (7)$$

Substitution of Equation (7) into Equation (4) yields an estimate of g_{sw} provided that $T_{\text{a,in}}$, RH_{in} , u_{in} , $T_{\text{a,cham}}$, RH_{cham} , T_{leaf} , $T_{\text{a,out}}$, RH_{out} , P_{atm} , and g_{blw} are known. Accurately measuring each of these variables in the air stream is difficult, and thus assumptions may be employed to close the mathematical equations.

2.1 | LI-COR LI-600's Approach to Closure

The LI-COR LI-600 porometer (LI-COR Inc., Lincoln, NE, USA) measures 5 of the 10 unknown input variables as indicated in Figure 1 (note that g_{blw} is not directly measured but is inferred based on other measurements and considered “measured” in our analyses).

The LI-600 does not measure chamber relative humidity, RH_{cham} , nor the different air temperatures throughout the air stream, $T_{\text{a,in}}$, $T_{\text{a,cham}}$, $T_{\text{a,out}}^{\text{direct}}$, and $T_{\text{a,out}}$. The design of the instrument is intended to minimise air temperature variation throughout the air stream. The inlet and outlet air streams are passed through an aluminium block manifold with high thermal conductance, and an internal fan forces air across the internal components to equilibrate temperatures. The temperature of the aluminium block is measured via a thermistor, and is called T_{ref} , and the heat exchange between this element and the chamber outflow results in $T_{\text{a,out}}$, which is assumed to equal T_{ref} so as to simplify its use in the above equations. In reality, there exists an immediate chamber outlet air temperature, $T_{\text{a,out}}^{\text{direct}}$, resulting from the potential evaporative cooling of a transpiring leaf. The degree to which the heat exchange block can equilibrate the immediate air stream temperature before it contacts the relative humidity sensor is then governed by some heat exchange efficiency, $\mathcal{E}$. LI-600's internal calculations implicitly assume that $\mathcal{E} = 1$ which results in $T_{\text{a,in}} = T_{\text{a,cham}} = T_{\text{a,out}}^{\text{direct}} = T_{\text{a,out}} = T_{\text{ref}}$. Furthermore, it is assumed that $RH_{\text{cham}} = RH_{\text{out}}$.

Although this design appears to simplify model calculations by forcing an approximately constant temperature throughout the system, it introduces new unknown and important variables which should be incorporated in model calculations. These unknown variables are the heat flux transferred to the chamber by convection, and the immediate outlet air temperature from the chamber.

2.2 | Estimating $T_{\text{a,out}}^{\text{direct}}$ Using Psychrometrics

When water vapour is adiabatically added to unsaturated air, the associated air temperature is reduced. We can use an energy balance to calculate this expected change in temperature.

Applying an energy balance between the chamber inlet and outlet, the change in enthalpy across the chamber is equal to

the convective energy added to the chamber by the chamber walls and leaf. The energy balance can be written as

$$u_{\text{out}}h_{\text{out}} - u_{\text{in}}h_{\text{in}} = Q_{H,\text{cham}}, \quad (8)$$

where h_{in} and h_{out} are the enthalpy of the inlet and outlet moist air (J per mol moist air), and $Q_{H,\text{cham}}$ is the rate of sensible heat added or removed from the chamber air stream (W), with the outlet mass flow rate calculated as $u_{\text{out}} = u_{\text{in}} + sE$. The enthalpy is given by

$$h(T, RH) = (1 - W) \cdot c_{p,a}T + W(\lambda_w + c_{p,w}T), \quad (9)$$

where $c_{p,a}$ is the specific heat capacity of dry air at constant pressure ($\sim 29.14 \text{ J mol}^{-1} \text{ }^\circ\text{C}^{-1}$), W is the water vapour mole fraction of the moist air ($W = e_s(T)RH/P_{\text{atm}}$; mol water vapour per mol moist air), λ_w is the latent heat of vaporisation of water ($\sim 44,000 \text{ J mol}^{-1}$), and $c_{p,w}$ is the specific heat of water vapour at constant pressure ($\sim 33.5 \text{ J mol}^{-1} \text{ }^\circ\text{C}^{-1}$).

We do not have sufficient information from available instrument measurements to properly calculate $Q_{H,\text{cham}}$, which is driven by forced convection from the instrument internal fan. This heat flux is also a function of heat generated from instrument electronics (e.g., the flow pump), from the chamber walls if the instrument is heated above the ambient temperature by direct solar radiation, and the ambient air temperature. We assume that $Q_{H,\text{cham}}$ is proportional to the difference between the inlet air temperature and the chamber air temperature:

$$Q_{H,\text{cham}} = C(T_{a,\text{in}} - T_{a,\text{cham}}), \quad (10)$$

where C is the thermal conductance ($\text{W }^\circ\text{C}^{-1}$), which needs to be empirically calibrated.

One could also consider using the measured leaf temperature T_{leaf} in estimating the convective heat added to the chamber air. The measured T_{leaf} can be noisy in some cases (it uses an infrared thermometer), and did not seem representative of the temperature difference driving this convective transfer. Incorporating it appeared to degrade results substantially.

The diffused outlet temperature, $T_{a,\text{out}}$, is a result of the direct outlet temperature, $T_{a,\text{out}}^{\text{direct}}$, and the effective heat exchange,

$$T_{a,\text{out}} = T_{a,\text{out}}^{\text{direct}} + \mathcal{E} \cdot (T_{a,\text{in}} - T_{a,\text{out}}^{\text{direct}}), \quad (11)$$

where $\mathcal{E}$ is the heat exchange efficiency (dimensionless) of the aluminium block depicted in Figure 1. When the heat exchange efficiency is 1.0, the diffused outlet temperature is distinct from the direct outlet temperature but equivalent to the block and inlet temperatures ($T_{a,\text{out}}^{\text{direct}} \neq T_{a,\text{out}}$; $T_{a,\text{out}}^{\text{direct}} = T_{a,\text{in}} = T_{\text{ref}}$). When the efficiency is 0, the diffused outlet temperature is equivalent to the direct outlet temperature and distinct from the inlet and block temperatures ($T_{a,\text{out}}^{\text{direct}} = T_{a,\text{out}} \neq T_{a,\text{in}} = T_{\text{ref}}$). For our primary results, we also assumed a heat exchange efficiency of 1.0, which leaves the correction with only one degree of freedom based on the convective coefficient C . In

the [Supporting information](#), we explore the more general case of a heat exchange efficiency possibly less than unity.

2.3 | Closing Equation (4) Based on $T_{a,\text{out}}^{\text{direct}}$ Estimation

Combining Equations (4), (7), (8), and (11), we are left with three equations and five unknowns: $T_{a,\text{in}}$, RH_{cham} , $T_{a,\text{out}}^{\text{direct}}$, E , and g_{sw} . It is reasonable to expect that the air temperature and humidity transition smoothly between the inlet and outlet values across the leaf/chamber, and thus $T_{a,\text{in}} > T_{a,\text{cham}} > T_{a,\text{out}}^{\text{direct}}$ and $RH_{\text{in}} < RH_{\text{cham}} < RH_{\text{out}}$. To avoid performing a complex integral across the leaf disk, we assume that

$$T_{a,\text{cham}} = \frac{1}{2}(T_{a,\text{in}} + T_{a,\text{out}}^{\text{direct}}), \quad (12)$$

$$RH_{\text{cham}} = \frac{1}{2}(RH_{\text{in}} + RH_{\text{out}}). \quad (13)$$

With these assumptions, we reduce our system to three equations and three unknowns, and as a result we can compute $T_{a,\text{out}}^{\text{direct}}$, E , and g_{sw} based on available LI-600 direct measurements by numerically solving the following nonlinear system

$$(\text{vapor diffusion}) \quad 0 = E - g_{\text{sw}}(W_{\text{leaf}} - W_{\text{cham}}), \quad (14)$$

$$(\text{mass balance}) \quad 0 = E - \frac{u_{\text{in}}}{s}(W_{\text{out}} - W_{\text{in}})(1 - W_{\text{out}})^{-1}, \quad (15)$$

$$(\text{energy balance}) \quad 0 = E - s^{-1} \left(\frac{sQ_{H,\text{cham}} + u_{\text{in}}h_{\text{in}}}{h_{\text{out}}} - u_{\text{in}} \right) \quad (16)$$

where W is the water vapour mole fraction, defined as above, for each location in the flow path, with respect to the T and RH at that location. Importantly, W_{out} and h_{out} use the diffused chamber outlet temperature, $T_{a,\text{out}}$, because it occurs at the same location as the RH_{out} sensor. This is assumed to equal T_{ref} in the results presented below.

3 | Methods

3.1 | Porometer-IRGA Instrument Comparison

The LI-600 Porometer/Fluorometer and LI-6800 Portable Photosynthesis System (LI-COR Biosciences, Lincoln, NE, USA) were directly compared in this study based on their reported values of stomatal conductance using 300 leaf patches across 13 species of varying stomatal magnitudes. All measurements for direct comparison were taken between the hours of 8:00 to 15:00 in field conditions on plants in a crop production setting or in a semi-urban setting (U.C. Davis Arboretum) in Davis, California, USA.

Healthy, fully-expanded sunlit leaves were randomly selected for measurement. For each leaf, 1 two-sided LI-6800 measurement paired with 6 one-sided LI-600 measurements were

collected, 3 on each side of the leaf patch as demonstrated in Supporting Information Figure S1. The 3 LI-600 measurements on each side were averaged to reduce the potential biases due to stomatal patchiness (Mott and Buckley 2000). The porometer measurements on each side were summed to yield the total g_{sw} value that was compared against the LI-6800, and were also used to calculate a stomatal sidedness ($1 + \frac{\min(g_{sw, top}, g_{sw, bottom})}{\max(g_{sw, top}, g_{sw, bottom})}$) value (Supporting Information Figure S4).

The LI-6800 was used with the clear-top chamber (6800-12A), a 2×3 or 1×3 cm² aperture depending on leaf size, a flow rate of 500 $\mu\text{mol s}^{-1}$, a chamber overpressure ΔP of 0.1 kPa, a fan speed of 5000 rpm, no humidity control, and no temperature control. Upon clamping onto the leaf, g_{sw} was monitored until stable, which typically occurred after between 30 s to 2 min, at which point the measurement was logged. The LI-600 was operated in *Auto Mode*, and the flow rate was set to *High* (150 $\mu\text{mol s}^{-1}$). In the data set presented, LI-6800 measurements were always performed first, with LI-600 measurements immediately following and in the same footprint area of the LI-6800 chamber (Supporting Information Figure S1). A subset of measurements was also collected in which the LI-6800 and LI-600 measurements were alternated in different orders (data not shown), which did not result in any meaningful differences. All paired measurements were taken within a 2–3 min or smaller window, which is not enough time for stomata to substantially react to the prior instrument's measurement, as most of these species exhibited stomatal time constants (time to $\sim 63.2\%$ of its asymptotic value) between 15 and 30 min in response to environmental perturbations.

3.2 | Correction Parameter Calibration

To determine the optimal thermal conductance coefficient, C , the correction was applied to the data set collected in this study using 100 values of C between 0.001 and 0.2 $\text{W } ^\circ\text{C}^{-1}$ on a log scale. Four performance metrics were computed using the corrected LI-600 data against the LI-6800 for each parameter combination - coefficient of determination (R^2), slope, bias, and root mean square error (RMSE). Their equations are given by

$$R^2 = 1 - \frac{\sum_{i=1}^n (g_{sw, LI-600, i} - g_{sw, LI-6800, i})^2}{\sum_{i=1}^n (g_{sw, LI-600, i} - \bar{g}_{sw, LI-600})^2}, \quad (17)$$

$$\text{Slope} = \frac{\sum_{i=1}^n (g_{sw, LI-6800, i} - \bar{g}_{sw, LI-6800})(g_{sw, LI-600, i} - \bar{g}_{sw, LI-600})}{\sum_{i=1}^n (g_{sw, LI-6800, i} - \bar{g}_{sw, LI-6800})^2}, \quad (18)$$

$$\text{Bias} = \left(\frac{1}{n} \sum_{i=1}^n \frac{g_{sw, LI-600, i}}{g_{sw, LI-6800, i}} - 1 \right), \quad (19)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (g_{sw, LI-600, i} - g_{sw, LI-6800, i})^2}. \quad (20)$$

The chosen C parameter for the correction used throughout the analysis is the one that minimised the following equally-weighted multi-objective function:

$$O = \frac{1}{3} \cdot |\text{Slope} - 1| + \frac{1}{3} \cdot |\text{Bias}| + \frac{1}{3} \cdot \frac{\text{RMSE}}{\overline{\text{RMSE}}}, \quad (21)$$

where $\overline{\text{RMSE}}$ is the mean RMSE across all parameter combinations. The multi-objective function was chosen in an attempt to reduce the bias evenly across the data set while maintaining a low RMSE. As seen in Figure A1, these are competing goals.

3.3 | Additional Data Set of Kimura et al. (2025)

For more robust results, we included previously published data of Kimura et al. (2025) in addition to our own due to its similar quality and its direct comparison of the LI-6800 clear top chamber and LI-600 in a field setting. We used this data for the analysis of the biases, and then applied the correction with the calibrated thermal conductance, C , solely from the data set collected in this study. Thus, the Kimura et al. (2025) data set acted entirely as a validation set not used in model calibration.

4 | Results

The mean percent bias and RMSE of the two-sided LI-600 measurements relative to the LI-6800 measurements increased exponentially with increasing stomatal conductance (Figure 2, Table 1). Biases start from around -10% to 10% at conductance values between 0 and 0.2 $\text{mol m}^{-2} \text{s}^{-1}$, and reach as high as around 100% at conductance above 0.8 $\text{mol m}^{-2} \text{s}^{-1}$. The biases observed are similar in magnitude to previously published findings with other porometer instruments (Lavoie-Lamoureux et al. 2017; Toro et al. 2019; Murray et al. 2019; Batke et al. 2020). The biases also increased with decreasing ambient relative humidity (Figure 4b).

The psychrometric correction was able to reduce the mean bias between the 6800 and 600 measurements of this study's data set from 33.9% to 10.7%, reduce the slope of the best fit line between them from 1.86 to 0.908, reduce the RMSE from 0.459 to 0.126 $\text{mol m}^{-2} \text{s}^{-1}$, and increase the R^2 from 0.737 to 0.835 (Figure 3) upon calibration that minimises the multi-objective function described in Section 3.2. The correction performs similarly across the range of relative humidities of this data set (30%–60%, Figures 4 and S6) and across the range of stomatal conductances (Figures 4 and Supporting Information Figure S5) for which there are large differences in bias across both. The resulting correction was minimal above roughly 50% humidity, although there were relatively fewer points at these higher humidity levels (Figure 4).

Based on the collected field data, we found that a C value of 0.0072 $\text{W } ^\circ\text{C}^{-1}$ jointly minimises the chosen error metrics when $\mathcal{E} = 1$ (Figure A1). When allowing $\mathcal{E}$ to vary alongside C , a Pareto front of near-optimal parameter combinations emerges, with $\mathcal{E} = 1$ included in that front (Supporting Information Figure S7). Thus, $\mathcal{E} = 1$, or the assumption that the air stream flowing out of the chamber equilibrates with the aluminium block temperature before passing over the RH_{out} sensor appears to be a reasonable one.

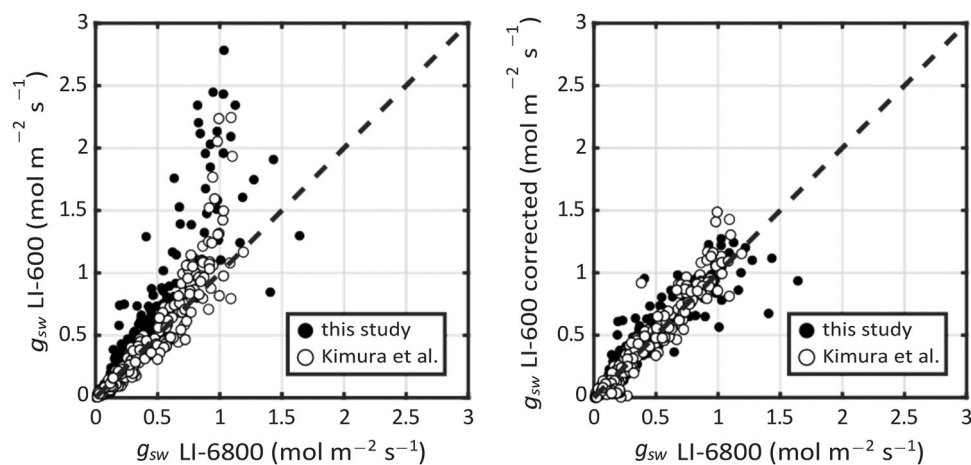


FIGURE 2 | Two-sided stomatal conductance measurements ($n = 610$) across 25 species using the LI-COR LI-6800 and LI-COR LI-600 porometer. Data is compiled from two datasets—this study (filled circles) and the work of Kimura et al. 2025 (open circles). Each point is one clear-top chamber LI-6800 measurement of a leaf section, followed by the sum of an average of 3 LI-600 measurements on each side of the same leaf section.

TABLE 1 | LI-600 stomatal conductance measurement bias, coefficient of determination, and root mean square error relative to the LI-6800 by conductance range among the two datasets.

Range ($\text{mol m}^{-2} \text{s}^{-1}$)	This study				Kimura et al. (2025)			
	n	Mean Bias (%)	R^2	RMSE ($\text{mol m}^{-2} \text{s}^{-1}$)	n	Mean Bias (%)	R^2	RMSE ($\text{mol m}^{-2} \text{s}^{-1}$)
(0.0, 0.2)	119	11.1	0.632	0.084	129	−13.3	0.818	0.027
(0.2, 0.4)	101	30.9	0.307	0.135	75	1.4	0.662	0.065
(0.4, 0.6)	35	34.8	0.022	0.250	32	−5.4	0.327	0.099
(0.6, 0.8)	13	64.3	0.034	0.535	38	10.1	0.460	0.185
(0.8, 1.0)	16	105.7	0.004	1.092	28	28.4	0.326	0.426
(1.0, ∞)	16	99.7	0.243	1.475	8	28.9	0.000	0.558
Overall	300	32.6	0.739	0.456	310	−1.2	0.864	0.176

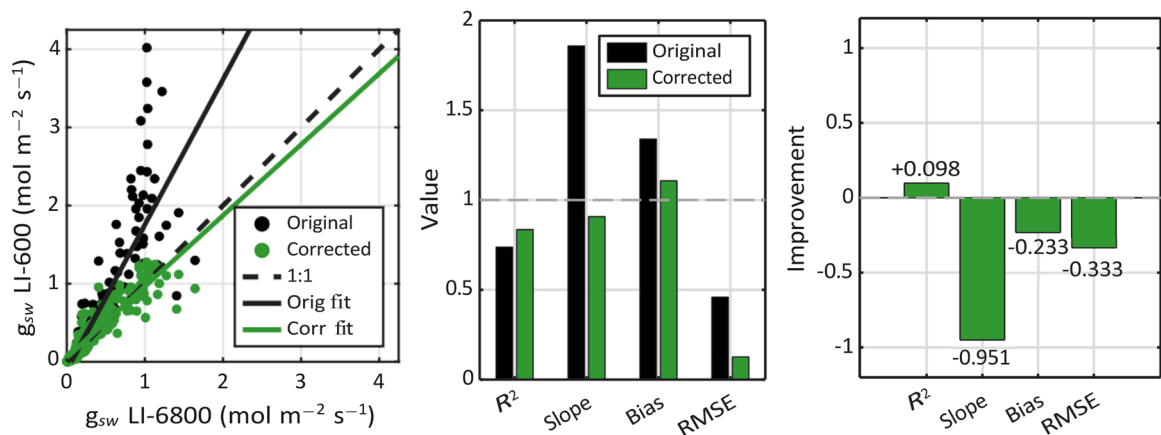


FIGURE 3 | The proposed psychrometric correction applied to two-sided measurements of the LI-6800 and LI-600 on the same leaves across 13 species measured in this study. The correction improves the coefficient of determination (R^2), the slope of the fit between the two instruments, the bias, and the root mean squared error (RMSE) in $\text{mol m}^{-2} \text{s}^{-1}$. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

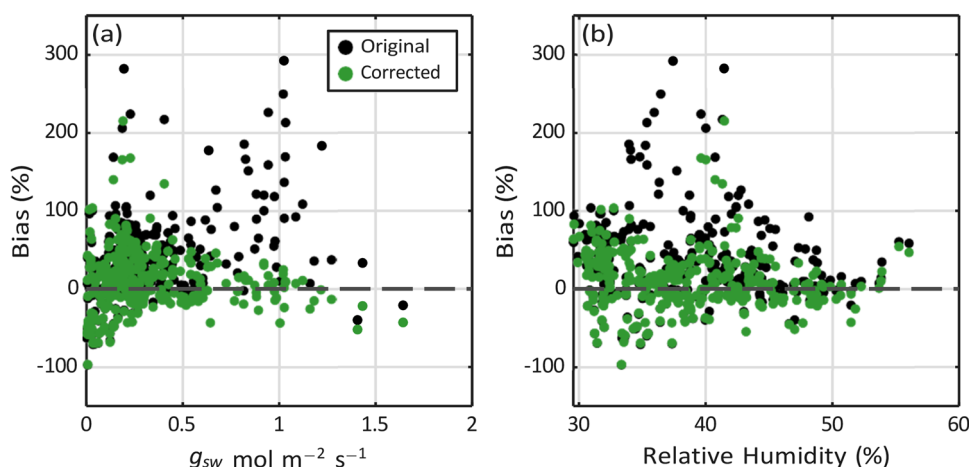


FIGURE 4 | The percent bias of the LI-600 stomatal conductance relative to the LI-6800 before (Original) and after (Corrected) the correction is applied to the data set of this study. The bias shows both a stomatal conductance (a) and relative humidity (b) dependence, with higher bias at higher stomatal conductance and lower relative humidities. The correction significantly reduces the bias across the range of both variables. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

5 | Discussion

5.1 | Absolute Versus Relative Stomatal Conductance

For many use cases in experimental plant physiological research and breeding trials, comparative measurements of stomatal conductance by porometer suffice as evidence to support a hypothesis or demonstrate statistically different conductances (e.g. Rebetzke et al. 2001; Wong et al. 2023; Buckley et al. 2025). However, when using stomatal conductance as an input to crop, land surface, or earth systems models, the accuracy of the absolute magnitude of conductance substantially impacts results. Additionally, comparing conductance values across studies and disciplines is hindered by the biases observed in porometry, eliciting a need for calibration between porometers and portable gas exchange systems (Toro et al. 2019).

5.2 | Psychrometric Versus Linear Correction

We found that applying a correction to measured stomatal conductance values that accounted for convective heat addition to the air stream and potentially air temperature changes throughout the system was able to correct the positive bias found in open flow-through porometric data (Figures 3 and Supporting Information S7). A few other studies have corrected porometer measurements with an entirely empirical approach (Turnbull et al. 2014; Mayanja et al. 2025) and one prior study also demonstrates a similar energy balance approach to theoretically improve porometry (Parkinson and Day 1980). The best linear correction that could be applied to our data set would appear to be a division of the observed g_{sw} value by 1.86. However, the psychrometric approach correctly applies minimal correction at low g_{sw} and moderate ambient humidity, and larger correction at high g_{sw} where the correction is needed (Figures 2 and Supporting Information S5). It also allows for further exploration into the causes of the bias and potentially informs the hardware in next generation systems. Furthermore, it suggests that there should be a measurable temperature gradient within the flow path given a transpiring

leaf, which could be validated experimentally with careful placement of thermocouples in various air stream locations.

5.3 | Empirical Thermal Conductance Calibration

While every term in the correction has a physical interpretation and could be further investigated, there does remain a somewhat unsatisfying degree of freedom in the thermal conductance, C (when $\mathcal{E}$ is fixed at unity). We took a multi-objective optimisation approach to empirically calibrate this parameter from data, but in theory, it could be further studied with thermal simulations and analyses of the instrument's internal parts.

Minimising the mean absolute bias and minimising the slope's deviation from one appear to occur in opposing directions with regard to the choice of C (Figure A1). Absolute mean bias reduction is important for aggregate bias reduction of the porometer and portable gas exchange system's reported stomatal conductance. Reduction in slope deviation from 1 promotes bias reduction across the whole range of conductance values, and is particularly important when large conductances are less frequent than small and moderate conductances. A solution exists (with a fixed $\mathcal{E}$ at unity) that entirely minimises the mean absolute bias of the data, but the resulting slope of the best fit line between the instruments in this case is around 0.7 (Figure A1), which is suboptimal. Similarly, RMSE has its own intermediate minimum. Thus, we chose to equally weight these objectives in the multi-objective function, which results in a higher remaining bias than is possible, but a slope closer to unity.

5.4 | A Variable Heat Exchange Efficiency and Computed Temperature Differentials

Relaxing the assumption of a perfect heat exchange efficiency, $\mathcal{E}$, of the aluminium heat exchange block, diagrammed in Figure 1, leads to an additional degree of freedom. We explored a space of possible C and $\mathcal{E}$ combinations in the [Supplementary Materials](#), Section S8.

The two-parameter correction was able to achieve better performance than the single-parameter correction, with the better bias, slope, and RMSE improvements— to 0.4%, 0.902, and $0.1211 \text{ mol m}^{-2} \text{ s}^{-1}$, respectively (Supporting Information Figure S8). Additionally, the temperature differences computed via the correction, $T_{a,\text{in}} - T_{a,\text{out}}^{\text{direct}}$, are more plausible under even a small reduction in $\mathcal{E}$, e.g., 0.84°C with $\mathcal{E} = 0.8$ compared to 4.01°C with $\mathcal{E} = 1$ (Supporting Information Table S2). Given this evidence, we suspect that an $\mathcal{E}$ of around 0.8–0.9 is reasonable, however, without further information regarding the true $\mathcal{E}$ or an independent method to estimate it, the correction with $\mathcal{E}$ fixed at unity provides a more parsimonious correction with similar results.

5.5 | Remaining Biases After Correction Varied Across Plant and Environmental Factors

After applying a correction to the datasets, the overall remaining bias was low, but varied across different environmental and plant factors (Figure A2). At high transpiration rates and high light intensities, the correction left little remaining biases. At low air temperatures ($<20^\circ\text{C}$) and low relative humidity ($<30\%$), the remaining percent bias is persistent. This could suggest an additional temperature-dependent mechanism that isn't being considered – the thermal conductance parameter, C , for example, could be temperature-dependent. However, it could also be a result of instrument precision limitations at very small conductances, which occur at these low air temperatures and low relative humidity.

Across locations (California and Japan) and across plant functional types (herbaceous and woody), the remaining biases were consistently low. However, individual species exhibited large variation in remaining biases. Given that the majority of these species were measured on only one or a few days, we suspect that this variation reflects the specific conditions of those days rather than intrinsic differences of the species. This interpretation aligns with the fact that the correction is founded on physical principles relating more to the instrument's physical properties and dynamics rather than that of the plant.

6 | Conclusion

The assumption of constant temperature throughout the air stream adopted by the LI-600 open flow-through porometer likely results in a consistent positive bias in computed stomatal conductance relative to the LI-6800 infra-red gas analyser-based system. By estimating the chamber and outlet air temperatures using a psychrometric energy balance approach, this bias was nearly eliminated in the data set used for the calibration and, importantly, in an entirely independent data set with different species not used for the calibration. The biases observed between the LI-600 and LI-6800 were not uniform across the observed range of stomatal conductance and relative humidity, which warrants the use of the proposed nonlinear psychrometric correction over a simpler linear correction. At low stomatal conductance ($\leq 0.25 \text{ mol m}^{-2} \text{ s}^{-1}$) and high relative humidity ($\geq 50\%$), the correction was generally minimal. However at lower humidity and g_{sw} around $1.0 \text{ mol m}^{-2} \text{ s}^{-1}$, the measurement of the LI-600 could be up to 3 times larger than that of the LI-6800.

While this approach effectively reduced the nonlinear biases observed, it remains somewhat unsatisfying in that it requires empirical calibration of a thermal conductance coefficient to estimate heat added to the flow by instrument electronics and chamber walls. It is encouraging that the correction based on this calibration still held when applied to an unseen independent data set from different species and a different climate. This coefficient could, however, be further investigated with empirical experimentation as it represents a real physical variable.

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Conflicts of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request. Routines for amending LI-600 output files with computed corrections to stomatal conductance and intermediary variables have been written in R, Python, MATLAB, and as a graphical user interface, and are available for community use and development (<https://github.com/ktrizzo/LI-600-StomatalCorrection>).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Figure S1: LI-COR LI-6800 (left) and LI-600 (middle) measurements being performed on the same leaf patch, with marks on the leaf to demonstrate that the typical pattern of LI-600 measurements always within the LI-6800-sampled leaf patch area. All measurements in the dataset were sampled with the LI-6800 once and the LI-600 a total of 6 times, 3 on the top and 3 on the bottom, with each 3 averaged to represent each respective side. **Figure S2:** Two different LI-600 instruments were compared back-to-back in differing orders on the same leaf patches throughout a walnut (*Juglans regia*) canopy on a mix of sunlit and shaded leaves with a range of stomatal conductances ($n = 24$). There was a +12% bias observed in instrument 1, the instrument used in this study - a significant bias but not on the order of the +50% bias observed in porometry versus infrared gas analysis comparisons. **Figure S3:** The stomatal sidedness ($1 + \min(g_{sw,top}, g_{sw,bottom})$) as obtained by LI-600 porometer measurements on both sides sw,top $sw,bottom$ of each sampled leaf patch ($n = 610$) across 25 species, with 1.0 corresponding to fully single-sided (hypostomatous in each case here), 2.0 corresponding to fully amphistomatous, and the continuous spectrum existing between. **Figure S4:** Original and corrected data of the LI-600 porometer and the mean bias compared to the LI-6800 portable gas exchange system binned by stomatal conductance (g_{sw}). The correction works across a spectrum of conductance values to reduce the bias of the instrument, which increases as g_{sw} increases. These results were obtained with $C = 0.007$ and $E = 1$. **Figure S5:** Original and corrected data of the LI-600 porometer and the mean bias compared to the LI-6800 portable gas exchange system binned by relative humidity (%). The correction works across a range of relative humidity (30–60 here) values to reduce the bias of the instrument, which increases as relative humidity decreases. These results were obtained with $C = 0.007$ and $E = 1$. **Figure S6:** Differences in temperature between the measured inlet reference temperature and the computed direct outlet temperature of the correction. This temperature differential generates the bias in reported stomatal conductance. These results were obtained with $C = 0.007$ and $E = 1$. **Figure S7:** Correction performance as a function of two calibration parameters - the thermal conductance between the inlet and chamber air streams, C , and the heat exchange efficiency of the aluminum block, E , that acts to bring the direct chamber outlet air temperature, T direct in equilibrium with inlet air temperature, T_{ref} . **Figure S8:** The performance of the two-parameter correction on the dataset of this study. The mean bias was reduced from 33.9% to 0.4%, a further reduction than the single parameter correction was able to achieve, but at the cost of an additional degree of freedom. The calibrated thermal conductance that minimized the equally-weight multi-objective function described in the text, in this case, was $C = 0.06$, and the heat exchange efficiency was $E = 0.222$. **Figure S9:** The bias reduction of the two-parameter correction on the dataset of this study, across the range of conductances. The overall mean bias was reduced from 33.9% to 0.4%, a further reduction than the single parameter correction was able to achieve, but at the cost of an additional degree of freedom. The calibrated thermal conductance that minimized the equally-weight multi-objective function described in the text, in this case, was $C = 0.0631$, and the heat exchange efficiency was $E = 0.2$. **Figure S10:** The bias reduction of the two-parameter correction on the dataset of this study, across the range of humidity. The overall mean bias was reduced from 33.9% to 0.4%, a further reduction than the single parameter correction was able to achieve, but at the cost of an additional degree of freedom. The calibrated thermal conductance that minimized the equally-weight multi-objective function described in the text, in this case, was $C = 0.0631$, and the heat exchange efficiency was $E = 0.2$. **Table S1:** LI-600 measurement bias relative to LI-6800 by species. **Table S2:** Optimal thermal conductance (C) for each heat exchange efficiency (E) with resulting metrics for both datasets.

Appendix A

List of Symbols Used

See Table A1

TABLE A1 | List of all symbols and descriptions for physical constants, correction parameters, measured variables, intermediately calculated variables, and solved-for variables.

Symbol	Description	Type	Value/Range	Units
Physical constants				
$c_{p,a}$	Heat capacity of dry air	Const	29.14	$\text{J mol}^{-1} \text{K}^{-1}$
$c_{p,w}$	Heat capacity of water vapour	Const	33.5	$\text{J mol}^{-1} \text{K}^{-1}$
λ_w	Latent heat of vaporisation	Const	45502	J mol^{-1}
g_{bhw}	Boundary layer conductance to water vapour	Const	2.921	$\text{mol m}^{-2} \text{s}^{-1}$
s	Leaf area	Const	0.441786×10^{-4}	m^2
Correction parameters				
C	Thermal conductance	Param	0.001–0.1	$\text{J s}^{-1} \text{K}^{-1}$
$\mathcal{E}$	Heat exchange efficiency	Param	0.0–1.0	—
s_{gs}	Stomatal sidedness	Param	1.0–2.0	—
Measured variables				
T_{ref}	Reference temperature of heat exchange block	Input	—	$^{\circ}\text{C}$
T_{leaf}	Leaf temperature	Input	—	$^{\circ}\text{C}$
RH_{in}	Inlet relative humidity	Input	0–1	—
RH_{out}	Outlet relative humidity	Input	0–1	—
u_{in}	Air flow rate	Input	—	mol s^{-1}
P_{atm}	Atmospheric pressure	Input	—	kPa
Calculated variables				
$T_{a,\text{cham}}$	Chamber temperature	Calc	—	$^{\circ}\text{C}$
$T_{a,\text{out}}$	Diffused outlet temperature	Calc	—	$^{\circ}\text{C}$
$T_{a,\text{in}}$	Inlet temperature (assumed = T_{ref})	Calc	—	$^{\circ}\text{C}$
RH_{cham}	Chamber relative humidity	Calc	0–1	—
e_s	Saturation vapour pressure	Calc	—	kPa
W	Water vapour mole fraction	Calc	—	mol water per mol moist air
h	Enthalpy	Calc	—	J mol^{-1}
g_{fw}	Total conductance to water vapour	Calc	—	$\text{mol m}^{-2} \text{s}^{-1}$
$Q_{H,\text{cham}}$	Sensible heat flux in the chamber	Calc	—	J s^{-1}
Solved variables				
$T_{a,\text{out}}^{\text{direct}}$	Direct outlet temperature	Output	—	$^{\circ}\text{C}$
E	Transpiration rate	Output	—	$\text{mol m}^{-2} \text{s}^{-1}$
g_{sw}	Stomatal conductance to water vapour, one-sided	Output	—	$\text{mol m}^{-2} \text{s}^{-1}$
$g_{sw,\text{total}}$	Stomatal conductance to water vapour, two-sided	Output	—	$\text{mol m}^{-2} \text{s}^{-1}$

Multi-Objective Optimisation

We explored how several error metrics and their weighted sum varied along choices of parameter values of the proposed correction applied to the data set collected in this study. The heat exchange efficiency, $\mathcal{E}$, was fixed at 1, and the heat transfer coefficient, C , was varied across 100 log-spaced increments from 0.001 to 0.2 $\text{W } ^\circ\text{C}^{-1}$. The error metrics (Equations (17–20)) and their weighted sum (Equation (21)) are shown across choices of C , each having a unique optimum (Figure A1).

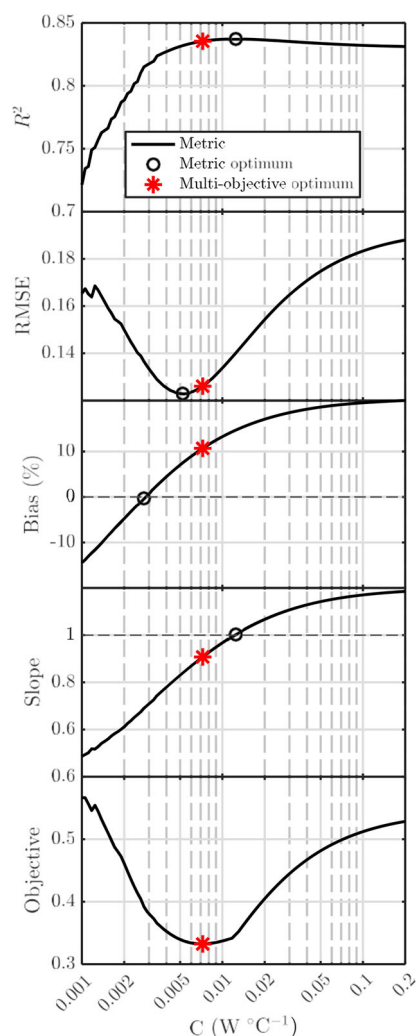


FIGURE A1 | Optimisation of empirical heat transfer coefficient, C , based on field data. When the heat exchange efficiency, $\mathcal{E}$, is assumed to be 1, the psychrometric correction is parameterised by one degree of freedom—the thermal conductance, C , between the inlet air and chamber air streams. Four error metrics were evaluated across a range of C values. The open black circle in each subplot denotes the optimal parameter values for that metric. The red star in each subplot denotes the multi-objective optimal values, which was comprised of an equal weighting ($\frac{1}{3}$) of three metrics: normalised RMSE, bias, and slope deviation from 1. The optimal C obtained from the multi-objective function was $0.0072 \text{ W } ^\circ\text{C}^{-1}$. This calibration was performed only to the data set collected in this study, then the resulting correction applied to both datasets for evaluation. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

Residual Biases

We applied the correction with parameters $C = 0.0631$ and $\mathcal{E} = 0.4$ to both datasets and explored the relationships between the pre-corrected and post-corrected biases against certain plant and environmental factors (Figure A2). Biases were computed with Equation (19), converted

to percent, and plotted against air temperature, relative humidity, light intensity, transpiration rate and VPD obtained from onboard LI-600 measurements. Plant type, location, and species are also highlighted. These specific correction parameters were chosen because they minimised the sum of the mean percent biases of each data set across our two-parameter search space (Supporting Information Table S2).

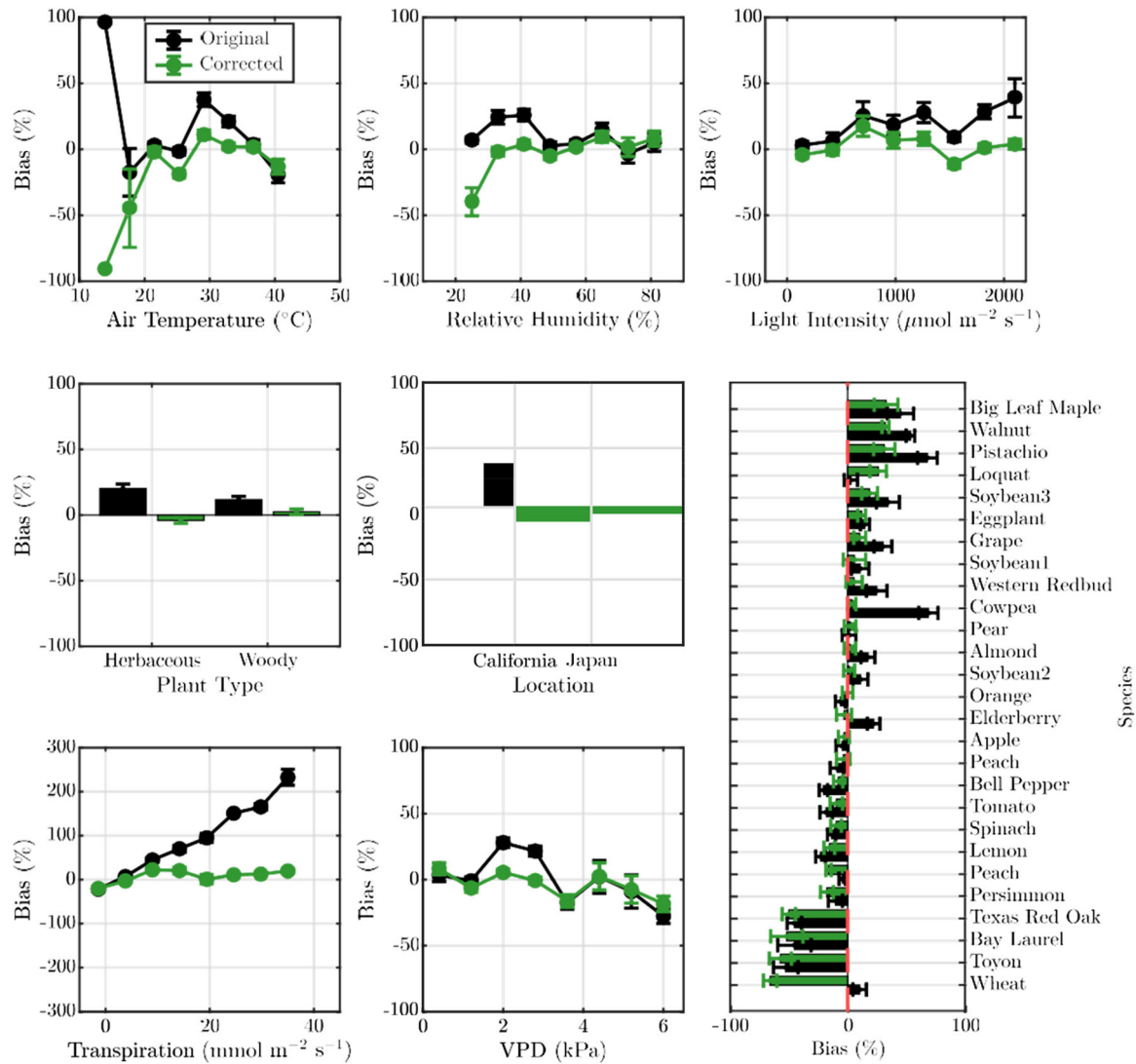


FIGURE A2 | Pre- and post-correction biases of stomatal conductance by porometry across multiple environmental and plant factors. After the two-parameter correction of $C = 0.0631 \text{ W}^\circ\text{C}^{-1}$ and $\mathcal{E} = 0.4$, which was the best found solution to jointly correct both datasets, the residual biases between stomatal conductance of the LI-600 and LI-6800 are plotted against air temperature, relative humidity, incident light intensity, plant type, geographic location, transpiration, VPD, and species. Note the difference in bias axis limits between those of transpiration and all other subplots. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]